# Chapter 1

# Literature Review

## 1.1 Theoretical Motivations and Background

This thesis belongs to a lineage of research in the field of language and gender. Scholars of language and gender look to the gendered indexicalities of everyday linguistic behavior as a medium for gender performativity. For the bulk of this thesis, I follow the tendency of language and gender scholarship in centering indexicality as the primary framework of analysis. Although performativity is the point of departure for much of language and gender research – especially following the importance of performativity within gender studies after the foundational work of Judith Butler (1990) (see also Bucholtz 2014) – indexicality provides a number of theoretical tools, as reviewed below, which make for a more actionable lens of analysis (Henry and Hall 2025). Sociolinguists study the "social-indexical performativity of language-in-use" (Silverstein 2023, p. 17) considering performativity to be mediated by indexicality (Henry and Hall 2025). For this reason, I move forward by investigating indexicality. In this chapter, I first introduce recent approaches to indexicality in language and gender research, demonstrating the turn towards an approach which considers speaker and listener together, and underlining theoretical questions which still remain. I then review the language and gender literature on /s/, arguing that the existing scholarship is unable to provide complete answers to these theoretical questions.

Indexicality comes to sociolinguistics through semiotics, the study of meaning-making. In this thesis I adopt the model of semiotics put forth by Gal and Irvine (2019), who synthesize prior approaches, placing critical emphasis on the procedural aspect of meaning making by which phenomena come to be meaningful. In their model, individuals make meaning of their world through conjectures. Drawing upon their knowledge and understanding of the world, they recognize a phenomenon and conjecture that it may be a sign, i.e. it may be meaningful. They conjecture that the sign signals (means or represents) a particular object, its meaning. Essential to note here is that these conjectures constitute the sign-object relationship; nothing is a sign in itself, independent of a perceiver conjecturing that it is a sign and it signals an object (Gal and Irvine 2019, p. 89). Also important is

that these conjectures are based on the perceiver's knowledge and understanding of the world, and which are constructed and mediated by their cultures, communities, and individual social identities– their lived experience (Gal and Irvine 2019; see also Calder 2021). Part of a conjecture is its "grounds": the reasoning the conjecture has as to why its phenomenon is a sign and why it signals this object. A conjecture is considered to be 'indexical' if its grounds are the knowledge or belief that the sign often co-occurs with, points to, or is caused by its object (Gal and Irvine 2019, p. 96).

Sociolinguistics often focuses on finding correlations between differences in language use (eg. pronunciation, grammar, word choice) and differences in social identity, persona, or context (Eckert 2012). Particular linguistic forms are then said to index the social identities of those who use them, on the grounds of this relationship of co-occurrence (Eckert 2012). As discussed using the example of voice pitch above, if a speaker uses a particular linguistic variant, and the listener believes that form to be commonly used by those of a particular gender, they may make a conjecture on the grounds of this co-occurence, considering the use of that form to index that speaker's gender. That there are two actors in play here, a speaker and a listener, led Bucholtz and Hall (2005; 2004; 2010) to understand the indexical-performative construction of gender (and identity categories more broadly) as an intersubjective process.

Bucholtz and Hall characterize this intersubjective process as a negotiation between the indexical goals of the speaker and the indexical goals of the listener (eg. Bucholtz and Hall 2005, p. 606). Although it is possible that a listener may construe an indexical meaning from a linguistic form used by a speaker who has no knowledge of, or intent to make use of, that form's indexicality, agentivity or intentionality on the part of the speaker is common (whether conscious or unconscious). Speakers have interactional goals, and hope that their behavior will be construed as indexical of those goals, whether it be related to identity, eg. presenting feminine, or other aspects such as persona and stance, eg. seeming like a 'diva' (Podesva 2007). This agentive view arose principally in reaction to earlier sociolinguistic models which stripped speakers of agency and individuality, instead understanding their linguistic behavior simply as a product of speakers' static, essentialized identities (Eckert 2012). In practice, this turn reframes "she speaks this way because she is a woman" as "she chooses this form to index femininity." Acknowledging that individuals have choices about how to present themself and granting them authority over their self-identification is also a move towards alignment with queer and trans understandings of identity; trans people in the US often consider gender identity to be self-determined (see Zimman 2017b), rather than a consequence of bodily characteristics. Despite the value of emphasizing the speaker's goals to legitimize their identities and agency, it is nonetheless the case that the listener is the powerful other side of this negotiation.

Inoue (2006; 2003) argues that in cases where the listener occupies a privileged social position of power over the speaker, they may inhabit a privileged 'listening subject' position in which the beliefs required to uphold this social hierarchy may play a primary role in their construal of that speaker's indexical behavior and

override or erase the speaker's intentions. Studying the emergence of the notion of 'women's language' in Japan, Inoue finds that to Meiji period elite male intellectuals, schoolgirls inhabited a social position which represented a modernizing breach from traditional Japanese social structure, and a threat to their power. For this reason, when they overheard schoolgirls using the verb ending forms *teyo* and *dawa*, they construed these endings as indexical of degeneracy and social decline. Of course, schoolgirls themselves did not intend to use these forms to index a 'degenerate' identity; the purported degeneracy these forms were said to represent was a social fabrication rather than a truth about any actions these girls were involved with. In this way, male intellectual overhearers constructed a listening subject position which made negative judgmental conjectures about the indexicality of schoolgirls' speech, and constructed a fictional object for the these speech forms to signal. The male listening subject sets the narrative, erasing schoolgirl speakers' semiotic agency.

Following this important insight, research has extended theorization of the listening subject to consider race and queerness, researching white (Flores and Rosa 2015) and cisgender listening subjects (Calder 2021; Heffernan 2023; Jessee and Calder 2025; Calder and A. Steele 2025). Flores and Rosa (2015) find that when white listeners racialize speakers as non-white, they justify their position of power over those speakers by construing those speakers' speech as indexing an underspecified 'inappropriateness' for the situation (eg., the classroom), regardless of those speakers' adherence to linguistic norms for standard English, constructing a white-supremacist white listening subject which will always deem non-white speech deficient and indexical of a supposed non-white inferiority. Jessee and Calder (2025) find that cisgender listeners apply stricter gendered listening standards to an audio clip of a voice when they are told the speaker of the clip is a trans woman than when told the speaker is a cis woman, similarly situating sufficient gender performance out of reach for transgender speakers in a move to deny their identities.

Drawing on these concepts of white, cis, and male listening subjects, Calder (2021) and Calder and A. Steele (2025) raise an important critique of the sociolinguistic project, questioning which listening subjects researchers themselves may be inhabiting in their interpretation of speakers' language use. Following the conjecturing model of Gal and Irvine (2019), they note that researchers collect numerical correlation data between linguistic forms and social aspects, and use this as the knowledge on which to base their conjectures of indexical links. Incorporating the research on the listening subject, they conclude that although this data may seem objective, empirical, and scientific, purely describing the world as it is (see also Zimman 2021a), listening practices inevitably are biased by the sociocultural positioning of the researcher and the scholarship that researcher builds upon. This critique raises the question of how to research the social-indexical performativity of language-in-use without taking up and reinforcing violent listening practices and marginalizing speakers who are not white, cis, heterosexual, middle class men. Work which addresses this question employs advanced models of indexicality which encourage sociolinguistic study to move from simply asking what

meanings participants index to investigating the socio-cultural structure of individuals' indexical knowledge, and the intersubjective process of indexicality (Calder 2021; Silverstein 2003; Silverstein 2023; Barrett and Hall 2024). The concepts of the indexical field as adapted by Calder (2021), and indexical (mis)recognition as developed by Barrett and Hall (2024) have been used to model indexical knowledge and the relativity and intersubjectivity of indexicality.

The concept of the indexical field, introduced by Eckert (2008), has been adopted by Calder (2021) to capture the way in which a particular phenomenon may be conjectured to have a multiple meanings, both within a single group of people and across different groups of people. The concept of the indexical field emphasizes, connecting to Gal and Irvine's (2019) semiotic model, that indexical relationships are conjectured through listeners' knowledge. In this way, a particular linguistic form may be construed as a sign of any number of meanings, depending on which knowledge the listener uses to conjecture the relationship, and what conclusion they come to in conjecturing. Although this knowledge is mediated and constructed differently by each individual's social and cultural lived experience, it is nonetheless to a large extent shared. For example, Inoue's (2003) listening subject was not a single Meiji-era Japanese male intellectual, it was an ideological position constructed and inhabited by this male intellectual social class. Individuals who inhabit similar social and cultural backgrounds and social positions, as well as co-members of communities, are likely to draw upon a similar base of knowledge, and are likely to consider certain phenomena signs with a particular set of possible meanings. Calder (2021) conceptualizes the indexical field as this set of meanings which may be shared by a set of people with shared epistemology. By making explicit whose indexical field researchers are investigating and interpreting signs through, the fact that this interpretation is coming from particular perspectives is forefronted, preventing the universalization of the hegemonic indexical field.

While this makes progress towards a more critical and less biased move to interpreting signs, and much research has been productively done to investigate the indexical fields within particular local communities, especially community-specific systems of categorization and meaning, it is nonetheless untenable to simply split the world into non-overlapping spheres of shared knowledge and shared indexicality. Individuals bring their own indexical knowledge into interaction, and are likely aware of a number of indexical fields. The question that arises is: what happens when listeners apply an indexical field differently than a speaker intends them to, or when there is a mismatch in indexical knowledge?

Barrett and Hall (2024) model these situations using their concepts of the indexical inventory and indexical (mis)alignment/(mis)recognition. For them, an individual's indexical knowledge is structured in an indexical inventory, which is "an individual's full set of indexical fields." (p. 132) When individuals interact and engage in sociolinguistic meaning-making through shared indexical meanings, they are said to be in indexical alignment. In this case, if a speaker uses a linguistic form that they know to hold a particular indexical meaning within one of the indexical fields in their inventory, their listener would recognize (in Barrett and

Hall's term) their behavior as indexical of the meaning on which they are aligned, interpreting it in line with the speaker's intentions. However, in cases of indexical misalignment, the speaker's intended interpretation does not come through: the speaker employs a linguistic form with the intent to index a particular meaning in their inventory, but the listener does not interpret the form as indexing that meaning. Although it is possible the listener may simply be unaware of the indexicality the speaker intended to make use of, Barrett and Hall (2024) emphasize that misrecognition is often politically motivated, whether this motivation be resistance against, or reproduction of hegemonic power.

For the purposes of this thesis, it is most relevant to review the marginalizing side of misrecognition, by which misinterpretation of signs is a tool for the reproduction of power. Barrett and Hall (2024) raise the example of the transition of generic language in English from masculine (eg. *he/mankind*) forms to gender neutral forms (eg. *they/humankind*). They note that in this transition, feminists constructed a new indexical meaning around generic masculines, considering these forms an index of misogynistic views in those who use them. In reaction, anti-feminists deny this new indexicality, considering generic masculines grammatical fact and misrecognizing, or refusing to recognize, new meanings. Here, Barrett and Hall consider the anti-feminist approach to be an example of their concepts of indexical presumption – whereby those in power presume that their indexical inventory is universal and correct – and indexical denial – in which knowledge of indexicalities against one's interests are denied.

When considered together with the concept of listening subjects, indexical presumption and indexical denial emerge as tactics which privileged listening subject positions might employ to in interpreting meanings from linguistic behavior. Taken as a whole, the concepts of the indexical field, listening subjects, and tactics of indexical misrecognition do the work of, in Barrett and Hall's terms, relativising indexicality. This makes it no longer tenable for researchers of sociolinguistic indexicality to draw correlations and make their interpretations from there; instead, they must work to understand the procedure by which meaning arises and all of the factors which influence speakers' choices in language use. In particular, this thesis is interested in asking how marginalized speakers deal with these listening practices, hoping to contribute to an understanding of how speakers model listeners' indexical inventories and how they make use of these models in making (conscious or unconscious) linguistic choices?

Although all speakers may to some extent model their listeners' indexical inventories, I choose to focus on the case of marginalized speakers as a group who is under the most pressure to do so. Barrett and Hall (2024) note that indexical meanings are not evenly distributed, and I would add that the work of modeling listeners' indexical inventories is also not evenly distributed. When marginalized speakers interact outside of safe communities, they are at risk of being judged negatively by listening subjects which will presume the hegemonic indexical field and deny alternate indexicalities. Conversely, privileged speakers are unlikely to face any negative consequences due to ignorance of indexical fields belonging to marginalized groups; marginalized listeners do not have the power to inflict the

same violence as priviliged listeners might. In reaction to this situation of imbalance, marginalized speakers are coerced to protect themselves by learning hegemonic indexicalities and predicting how listeners may interpret their speech from an antagonistic listening subject position.

Heffernan (2023) reveals that transfeminine speakers are aware of the cisgender listening subject. Her study collected metalinguistic commentaries on transfeminine individuals relationships with their voices and experiences with being heard from an audio-recorded conversation she conducted between two US (South Asian American and White American) trans women and from anonymous transfeminine users on the Reddit online forum r/TransVoice. Commentaries see that cisgender listeners hear transfeminine voices through hegemonic norms and make binary judgements, categorizing voices as belonging to a man or a woman. Transfeminine speakers do not want to sound like men, and are aware of the possibility that cisgender listeners may make this judgements. In these commentaries, speakers additionally discuss what they do with this awareness and how they respond to this risk, a question I take as central to my study.

Heffernan finds that speakers simultaneously express feelings of pressure to modify their voices to avoid these negative judgements and feelings that they should not have to do so, exposing the conflict between "views of gender as self-determined, on the one hand, and of gender as intersubjectively constituted, on the other" (ii). These two approaches can be found in the commentaries from one of the audio-recording participants. This participant notes that when "interacting with the cis universe" she finds herself "deliberately modulating [her] voice" to make it "more girly," and that this is "a kind of survival tactic... to try and avoid any possibility that [she]'ll get misgendered" (23). In this way, the speaker acknowledges the risk of misgendering that comes from interacting with cisgender listeners, and her response is to change her language use in an attempt to align more closely with hegemonic feminine norms. However, she also asserts that "it's kind of just always incorrect to assume someone's gender," which and that she is uncertain about whether she would want to train herself to always use a more hegemonically feminine voice, as she wants to be able to "[think] about (.) [her] voice ... through uh- like the lens of other trans people", who would not apply the hegemonic indexical field, not assume her gender, and who would accept her voice as it is. This tension is around the decision to shift one's speech in the presence of a privileged listener who may inhabit a judgmental listening subject position is of primary interest to my study. Because she only investigated metalinguistic commentaries, Heffernan was not able to investigate how these decisions play out in transfeminine people's real interactions with cisgender listeners; this thesis aims to fill this gap.

Within the existing literature, I identify three sutdies which have considered changes in individual transgender speakers' language use based on context: Becker, Khan, and Zimman (2022), Gratton (2016), and Parnell-Mooney (2019). Becker, Khan, and Zimman (2022) find that trans men and nonbinary people assigned female at birth on testosterone use significantly less creaky voice (known popularly as 'vocal fry') in a reading-passage task compared to in a open-ended conversation.

As creaky voice is associated with femininity within the North American English hegemonic indexical field, they tentatively interpret these findings as evidence that in a more scrutinized form of speech (in which, I add, the notion of the cisgender listening subject may be more mentally present), these individuals purposefully move away from normative feminine speech in order to assert their identities as non-women.

This aligns with the findings in a study on a Toronto nonbinary friend group conducted by Gratton (2016). Gratton finds that variation between *-ing* and *in'* (eg. *working* vs. *workin'*) was more even in queer-safe spaces, where speakers used roughly the the same amount of both, and more differentiated in public, where speakers preferred one form over the other. In the hegemonic indexical field, *-ing* variant is hegemonically associated with femininity and *in'* with masculinity. Gratton finds that in public, speakers avoid the variant which is hegemonically indexical of their assigned gender at birth, understanding that cisgender overhearers are invested imposing that gender onto them and choosing to exaggerate their difference from that gender to avoid that perception. In queer-safe spaces, (in which, I add, the cisgender listening subject is not present), speakers were free to speak more neutrally and without regard for binary norms – Gratton notes that one speaker assigned male at birth mentions feeling comfortable making occasional use of a 'manly voice' in these spaces, as they know they will not face misgendering for doing so in the absence of cisgender listeners.

Parnell-Mooney (2019) also finds this sort differentiation, finding that in conversations with cis listeners, four transmasculine and two nonbinary speakers shift to pronounce /s/ further from the type of pronunciation expected of their assigned gender at birth. The trans men and the nonbinary person assigned female at birth shift further from hegemonically feminine /s/ pronunciations with with cisgender women listeners than when interacting with cisgender men, whereas the nonbinary person assigned male at birth shifts the opposite way. Although these studies present important empirical results, they do not integrate with the theoretical material discussed above – listening subjects, indexical fields, indexical recognition, and the modeling of a listener's indexical knowledge. They also do not include transfeminine speakers, and do not explicitly compare speech to a cis listener against speech to a trans listener. Filling this gap, I compare transfeminine speakers' indexical behavior across a conversation with a cisgender woman and a transgender woman with reference to their commentaries on their own gender, their voice, and being heard as a transfeminine speaker.

For my comparison, I focus specifically on the English /s/ sound due to the richness of existing research on its indexicalities and its use by queer and trans speakers, especially in connection to the sophisticated theories of indexicality discussed above. The remainder of this section introduces the literature on /s/ and reviews its connections to the concerns of transgender language and indexicality research.